Assignment_Linear Transformations

1. Let V and W be vector spaces, let T: V → W be linear, and let $\{w_1, w_2, \ldots, w_k\}$ be a linearly independent subset of R(T). Prove that if $S = \{v_1, v_2, \ldots, v_k\}$ is chosen so that $T(v_i) = w_i$ for $i = 1, 2, \ldots, k$, then $S$ is linearly independent.

2. Let V and W be vector spaces and T: V → W be linear.
**(a)** Prove that T is one-to-one if and only if T carries linearly independent subsets of V onto linearly independent subsets of W.
**(b)** Suppose that T is one-to-one and that $S$ is a subset of V. Prove that $S$ is linearly independent if and only if $T(S)$ is linearly independent.
**(c)** Suppose $\beta = \{v_1, v_2, \ldots, v_n\}$ is a basis for V and T is one-to-one and onto. Prove that $T(\beta) = \{T(v_1), T(v_2), \ldots, T(v_n)\}$ is a basis for W.

3. Let V and W be vector spaces, and let $\{v_1, v_2, \ldots, v_k\}$ be a linearly dependent subset of vectors of V. Suppose $\dim W \geqslant 1$. Show that there exists $\{w_1, w_2, \ldots, w_k\}$ of W such that there does not exists a linear transformation T: V → W such that $T(v_1)=w_1, v \ldots, T(v_k)=w_k$.

4. Let V be an $n$-dimensional vector space over a field K with an ordered basis $\beta$. Define
T: V → $K^n$ by $T(x) = [x]_\beta$, the coordinates of x w.r.t. $\beta$.

Prove that T is linear.

5. A function T: V → W between vector spaces V and W is called **additive** if $T(x + y) = T(x) + T(y)$ for all $x, y \in V$. Prove that if V and W are vector spaces over the field of rational numbers, then any additive function from V into W is a linear transformation.

6. Let V and W be vector spaces with subspaces $V_1$ and $W_1$, respectively. If T: V $\rightarrow$ W is linear, prove that T($V_1$) is a subspace of W and that $\{x \in V: T(x) \in W_1\}$ is a subspace of V.

7. Give an example of distinct linear transformations T and U such that Ker(T) = Ker(U) and Im(T) = Im(U).

8. Let V be a finite-dimensional vector space and T: V $\rightarrow$ V be linear.
**(a)** Suppose that V = Im(T) + Ker(T). Prove that V = Im(T) $\oplus$ N(T).
**(b)** Suppose that Im(T) $\cap$ Ker(T) = {0 }. Prove that V = Im(T) $\oplus$ Ker(T).

9. Let T: $R^2 \rightarrow R^3$ be defined by T($a_1$, $a_2$) = ($a_1 - a_2$, $a_1$, $2a_1 + a_2$). Let $\beta$ be the standard ordered basis for $R^2$ and $\gamma$ = {(1, 1, 0), (0, 1, 1), (2, 2, 3)}. Compute $[T]_{\gamma}^{\beta}$. If $\alpha$ = {(1, 2), (2, 3)}, compute $[T]_{\gamma}^{\alpha}$.

10. Let V be the vector space of complex numbers over the field R. Define T: V $\rightarrow$ V by T($z$) = $z$, where $z$ is the complex conjugate of $z$. Prove that T is linear, and compute $[T]_{\beta}$, where $\beta$ = {1, i}.

11. Let V be a vector space, and let T: V $\rightarrow$ V be linear. Prove that $T^2 = 0$ if and only if Im(T) $\subseteq$ Ker(T).

12. Let V, W, and Z be vector spaces, and let T: V $\rightarrow$ W and U: W $\rightarrow$ Z be linear.
**(a)** Prove that if UT is one-to-one, then T is one-to-one. Must U also be one-to-one?
**(b)** Prove that if UT is onto, then U is onto. Must T also be onto?
**(c)** Prove that if U and T are one-to-one and onto, then UT is also.

13. Let V be a finite-dimensional vector space, and let T: V $\rightarrow$ V be linear.
**(a)** If rank(T) = rank($T^2$), prove that Im(T) $\cap$ Ker(T) = {0 }. Deduce that V = Im(T) $\oplus$ Ker(T) .

**(b)** Prove that $V = Im(T^k) \oplus Ker(T^k)$ for some positive integer $k$.

14. Let V be a finite-dimensional vector space, and let $T: V \to V$ be linear. Show that there exists an integer m such that for any $k \geqslant m$ $Ker(T^k)=KerT^m$.

15. Let V be a vector space. Determine all linear transformations $T: V \to V$ such that $T = T^2$. *Hint:* Note that $x = T(x) + (x - T(x))$ for every $x$ in V, and show that $V = \{y : T(y) = y\} \oplus Ker(T)$

16. For each of the following pairs of ordered bases $\beta$ and $\beta_$ for $P^2(R)$, polynomials with degree less than or equal to 2,
find the change of coordinate matrix that changes $\beta_$-coordinates into $\beta$-coordinates.
**(a)** $\beta = \{x^2, x, 1\}$ and
$\beta_ = \{a_2x^2 + a_1x + a_0, b_2x^2 + b_1x + b_0, c_2x^2 + c_1x + c_0\}$
**(b)** $\beta = \{1, x, x^2\}$ and
$\beta_ = \{a_2x^2 + a_1x + a_0, b_2x^2 + b_1x + b_0, c_2x^2 + c_1x + c_0\}$

17. (i)Let V, W be vector spaces such that dimV=n, dimW=m. Let $T: V \to W$ , $T': V \to W$ be linear transformations such that KerT=KerT'. Show that there exists an isomorphism S of W such that T=ST'. (ii) Let A, B be mxn matrices. Such that Nul(A)=Nul(B). Show that there exists an invertible mxm matrix E such that A=EB.

18. (i)Let V, W be vector spaces such that dimV=n, dimW=m. Let $T: V \to W$ , $T': V \to W$ be linear transformations such that RankT=RankT'. Show that there exists an isomorphism of V and S of W such that T=ST'Q. (ii) Let A, B be mxn matrices. Suppose RankA=RankB. Show that there exists an invertible mxm matrix E and an invertible nxn matrix F such that A=EBF.

19. Let T: V $\rightarrow$ V be a linear transformation such that for any linear transformation S: V $\rightarrow$ V, we have ST=TS. Show that T is a multiple of the identity map i.e. T= aI for some number a.

20. Let V be a finite dimensional vector space. Suppose T, $S_1$, $S_2$ : V $\rightarrow$ V are linear transformations such that $S_1TS_2$ =I. Show that T is invertible and that $T^{-1}=S_2S_1$.

21. Let A be an mxn matrix and B be an nxr matrix. Show that

Rank(AB)$\leqslant$ min( Rank(A), Rank(B))

22. Let A, B be two mxn matrices. Show that

Rank(A+B)$\leqslant$Rank(A) + Rank(B)